

# Dr. Stefan Ellmauthaler

## Curriculum Vitae

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Dr. rer. nat. | AI Researcher | Tech & AI Consultant | NixOS Enthusiast | Speaker

### Employment History

- 12/2024 - today **CEO and Founder**, Echo Intelligence GmbH, Krenngasse 12, A-8010 Graz, Founded Echo Intelligence GmbH to deliver cutting-edge AI and IT services. Oversaw all legal and commercial aspects of the business. Officially recognized as a NIS2 consultant by UBIT Steiermark. Member of the IT Security Experts Group Steiermark and Austria.
- 01/2021 - 09/2024 **Postdoctoral Research Associate**, TU Dresden, Faculty for Computer Science / Center for Advanced Electronics Dresden (cfaed) / International Center for Computational Logic (ICCL), Knowledge-Based Systems Group, Research on qualitative AI, rule-based reasoning, ontology-based data access and argumentation theory. Rust software development, system administration, various organisational tasks, and teaching. Project management responsibilities as InnoSale's work package lead.  
Organiser of SAFA 2024, technical chair of JELIA 2023, and various reviewing activities.  
**Funded by** the Federal Ministry of Education and Research (BMBF) in ScaDS.AI (Center for Scalable Data Analytics and Artificial Intelligence) and additionally in the BMBF funded industry-oriented ITEA4 project "Innovating Sales and Planning of Complex Industrial Products Exploiting Artificial Intelligence (InnoSale)" since 01/2023.
- 10/2018 - 12/2020 **Postdoctoral Research Associate**, Leipzig University, Computer Science Institute, Intelligent Systems Group, State-funded position with obligatory teaching duties. Handling of COVID-19 related administrative and organisational tasks. Guest-Editor for TLPL. Various reviewing tasks.
- 10/2012 - 09/2018 **Research associate**, Leipzig University, Computer Science Institute, Intelligent Systems Group, Research on artificial intelligence and knowledge representation in distributed and stream-based reasoning, argumentation theory, and rule-based logic programming. Teaching, the organisation of multiple peer-reviewed workshops (RactKnow 2014, IULP 2015 and 2017) and various reviewing activities. Administrative and project management tasks for the Hybris-research group. Working at the following projects of the german research group (DFG) as **project employee**:
- 10/2017-09/2018: "Advanced tools for graph-based formal argumentation" (BR-1817/7-2)
- As **project employee** and **administrative officer** for Leipzig:
- 10/2012-09/2017 "Hybris-A2: Advanced Solving Technology for Dynamic and Reactive Applications" (Researchgroup Hybrid Reasoning for Intelligent Systems - "Hybris" - FOR 1513)
  - 10/2012-09/2015 "Hybris-B2: From Correlation to Causality: Hybrid Reasoning over Dynamic Protein Interaction Networks" (FOR 1513)
- 03/2011 **Project employee**, Vienna University of Technology

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## Education

- 10/2012 – 06/2018 **Doctoral study (Dr. rer. nat.)**, *Computer Science*, Leipzig University, “magna cum laude”.  
“Multi-Context Reasoning in Continuous Data-Flow Environments”  
• Prof. Dr. Gerhard Brewka (supervisor and first reviewer)  
• Prof. Dr. Torsten Schaub (second reviewer)
- 10/2009 - 09/2012 **Master's study (Dipl.-Ing.)**, *Computational Intelligence*, Vienna University of Technology, with distinction. “Abstract Dialectical Frameworks: Properties, Complexity, and Implementation”  
• Privatdoz. Dr. techn. Dipl.-Ing. Stefan Woltran (supervisor)  
• Proj. Ass. Dipl.-Ing. Johannes Peter Wallner, BSc (assistance)
- 10/2005 - 07-2009 **Bachelor's study (BSc)**, *Medicine and Computer Science*, Vienna University of Technology
- 09/2000 - 06/2005 **"Matura" Higher School Certificate**, *Department EDVO*, HTBLuVA Wiener Neustadt, School to gain matriculation standard in conjunction with expertise in the field of informatics and business management (graduation at the 13th school year)

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## Further Education

- 02/2026 **Updateseminar NIS-2: Netz und Informationssicherheitsgesetz 2026 (NISG 2026)**, UBIT Akademie Incite, WIFI Steiermark, Graz (online), Updateseminar of 2 teaching units about the current NISG2026 implementation of NIS-2
- 09/2025 **Certified Final Examination Assessor**, WIFI Steiermark, Graz, Training of 8 hours to become a certified final examination assessor
- 09/2025 **Basic Workshop Digital Consultant**, UBIT Akademie Incite, WIFI Steiermark, Graz (online), Workshop of 8 teaching units on the topics: Digitalization, Trend Maps, Digital Status and Potential Analysis, Maturity Assessment, Reporting Requirements and KMU.DIGITAL
- 05/2025 **Cybersicherheitsrichtlinie NIS 2**, UBIT Akademie Incite, WIFI Steiermark, Graz, Workshop of 10 teaching units on NIS 2. Legal aspects, risk management, measures
- 06/2024 **Business administration for young researcher**, Graduate Academy TU Dresden, Workshop of 16 teaching units on business administration, Lecturer: Prof. Dr. Torsten Forberg
- 05/2024 **Negotiating in Science & Academia**, Graduate Academy TU Dresden, Workshop of 9 teaching units on negotiations during business deals and academia, Lecturer: Dr. Silvia Scheinert
- 10/2023 **First Aid Training including Sports Emergencies**, Deutsches Rotes Kreuz Dresden, Course comprising 9 teaching units
- 03/2023 **Managing Projects and Leading Teams for Postdocs**, Graduate Academy TU Dresden, Workshop of 13 working units, Lecturer: Dr. Markus Schmittberger
- 05/2022 **Leadership Skills for Postdocs**, Graduate Academy TU Dresden, Workshop of 18 working units, Lecturer: Hiu Pun Chong
- 01/2022 **Agile Projectmanagement - Methods and Processes to Cooperation in Teams**, Center for continuing education - TU Dresden, Workshop of 18 working units, Lecturer: Roberto Riehle (SCRUM-Master and Agile-Coach)
- 10/2018 - 06/2020 **Saxon Higher Education Didactics Certificate**, Center for Didactics and Higher Education Saxony (HDS), 240 working units

- 07/2014 **Doctoral Consortium**, Fourteenth Conference on Principles of Knowledge Representation and Reasoning (KR), Wien, Topic: Argumentation for Reactive Reasoning
- 06/2013 **Summer School**, Advanced Course on AI (ACAI), London

## Skills

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|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Expert               | Knowledge Representation, qualitative AI, Argumentation Theory, Complexity Theory, Discrete Planning and Optimisation, LaTeX, typst, Higher Education Teaching, Scientific working methods, Research organisation, Declarative Programming (Answer Set Programming) |               |
| Further Competencies | (Agile) Projectmanagement, NixOs, bash/zsh, python, C(++), Rust, SQL, datalog, git, Basic Machine Learning (BDT, Deep Learning) knowledge, Basic Data Science knowledge                                                                                             |               |
| Softskills           | team player, adaptable, conflict and problem discussion experience, leadership skills, presentation skills, MBTI: ESTJ                                                                                                                                              |               |
| Languages            | German                                                                                                                                                                                                                                                              | native tongue |
|                      | English                                                                                                                                                                                                                                                             | fluent        |

## Awards

- 07/2017 **7. Answer Set Programming Competition Modelling Competition**, International Conference on Logic Programming and Nonmonotonic Reasoning (LPNMR) 2017, Espoo, Finland: *first place*
- 07/2014 **ASP Modelling Contest**, International Conference on Logic Programming (ICLP) 2014 during the Vienna Summer of Logic 2014 Olympics, Vienna, Austria: *third place*
- 08/2013 **20. PROLOG Programming Contest**, International Conference on Logic Programming (ICLP) 2013, Istanbul, Turkey: *first place*

## Volunteering Work

- 01/2026 - today **Chair of the Final Examination Commission** , WKO Steiermark  
*Examination commission chair for the final examination in the apprenticeship professions **Application Development - Coding and Information Technology - System Technology/Operational Technology**.*
- 09/2025 - 07/2026 **Head of IT & Event Technology at Ideentriebwerk**, Association for the Visibility of the Startup Community in Austria, Graz  
*Leading the IT team. Administration of IT infrastructure, organization and operation of event technology at events. Reporting to the core-team and the board.*
- 07/2025 - today **Member of the Final Examination Commission (Assessor; Chair since 01/2026)**, WKO Steiermark  
*Examination commission for the final examination in the apprenticeship professions **Application Development - Coding and Information Technology - System Technology/Operational Technology**.*
- 04/2025 - today **Ideentriebwerk**, Association for the Visibility of the Startup Community in Austria, Graz  
*Member of IT and event technology, administration of IT infrastructure, organization and operation of event technology at events.*

11/2023 - 09/2024 **Company First Aider**, TU Dresden

10/2023 - 09/2024 **DHSZ Course/Coach**, Dresden University Sports Center, TU Dresden  
*Planning and conducting a weekly table tennis meeting lasting 90 minutes for employees of the Faculty of Computer Science.*

## Academic Working Experience

### Reviewing

Journal Reviewing Theory and Practice of Logic Programming (TPLP), ACM Transactions on Database Systems (TODS), Artificial Intelligence Review (AIRE), IfCoLog Journal of Logics and their Applications, Journal of Artificial Intelligence Research (JAIR), International Journal of Approximate Reasoning (IJA)

Program Committee AAAI 2026, ASPOCP 2026, JELIA 2025, AAAI 2024, ASPOCP 2024, ECAI 2024, AAAI 2023, ASPOCP 2023, ECAI 2023, ASPOCP 2022, SAFA 2022, ASPOCP 2020, COMMA 2020, SAFA 2020, ASPOCP 2019, AAAI 2018, SAFA 2018, COMMA 2016, SAFA 2016, IJCAI 2015 (KR Track)

Reviewing Conferences & Workshops COMMA 2018, IJCAI 2018, IJCAI 2017, IJCAI 2016, ICLP 2016, SAFA 2016, IJCAI 2015, IULP 2015, AAAI 2014, ECAI 2014, ReactKnow 2014, ICCSW 2014, ICLP 2013, ICCSW 2013

### Scientific Working Experience

SAFA Organiser 5. International Workshop on Systems and Algorithms for Formal Argumentation (SAFA2024), co-located with the 10. International Conference on Computational Models of Argument (COMMA 2024) at Hagen, Germany

JELIA Technical Chair 18. Edition of the European Conference on Logics in Artificial Intelligence (JELIA 2023) at Dresden, Germany

KI Tutorial Speaker Tutorial on “Multi-Context Stream Reasoning” at the 43. German Conference on Artificial Intelligence (remote, 2020)

Guest-Editor Theory and Practice on Logic Programming (TPLP), Special issue on User-Oriented Logic Programming and Reasoning Paradigms

IULP Organiser 2. International Workshop on User-Oriented Logic Paradigms (IULP 2017), co-located with the 4. International Conference on Logic Programming and Nonmonotonic Reasoning (LP-NMR 2017) at Espoo, Finland

IULP Organiser International Workshop on User-Oriented Logic Programming (IULP2015), co-located with the International Conference on Logic Programming (ICLP 2015) at Cork, Ireland

ReactKnow Organiser International Workshop on Reactive Concepts in Knowledge Representation (ReactKnow 2014), co-located with the European Conference on Artificial Intelligence (ECAI 2014) at Prague, Czech

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## Teaching Experience

### Teaching Responsibilities

WT 2021, ST 2022    Transfer of teaching responsibilities on knowledge based systems at TU Dresden

### Lectures/Courses

- ST 2023, WT 2023    **CMS Team Project**, *Software Team Project*, 8.0h per week, 300 hours workload, Advanced MSc module course in computer science with software engineering and teamwork focus
- WT 2021    **Rule-Based Paradigms in Knowledge Representation**, *Seminar*, 2.0h per week, Advanced MSc module course in computer science
- ST 2015 - 2020    **Knowledge Based Systems**, *Exercise*, 1.0 h per week, BSc course in computer science, Lecture held by Prof. G. Brewka
- SS 2015 - 2020    **Knowledge Based Systems**, *Lecture*, 2.0 h per week, Represented Prof. G. Brewka at Topic "ASP", BSc course in computer science
- WT 2019    **Knowledge Representation**, *Lecture*, 2.0 h per week, Presented the topic "Multi-Context Systems", MSc course in computer science
- WT 2013, WT 2019    **Knowledge Representation**, *Seminar*, 2.0 h per week, MSc course in computer science
- WT 2017 - 2020    **Knowledge Representation**, *Practical course*, 2.0h per week, Learn on how to model knowledge with answer set programming, MSc course in computer science
- WT 2014    **Intelligent systems**, *Seminar*, 2.0 h per week, MSc course in computer science
- WT 2014    **Logic in Computer Science**, *Seminar*, 2.0 h per week, MSc course in computer science
- ST 2013    **Semantic Web Technologies**, *Seminar*, 2.0h per week, MSc course in computer science

### Co-supervised Bachelor's Thesis

- 2022    *Justus Möller*: Construction of Human-solvable Sudoku Puzzles
- 2020    *Tobias Krawetzke*: Dialectical Models Encoding in C++ neu betrachtet: Analyse, Benutzbarkeit und Modifikation, Grade: "good"
- 2017    *Kevin Kubsch*: Application and implementation of splitting procedures to Abstract Dialectical Frameworks, Grade: "very good"
- 2014    *Jakob Runge*: CARBON - a Web application and a RESTful API for argumentation

### Teaching Assistance

- 2006 - 2009    **Tutor**, Vienna University of Technology, Institute of Computer Aided Automation, Tutor for the BSc course "EProg" (Introduction to Programming)"